Real Numbers, Exponents & Radicals

Notes & Practice Worksheet

FANG MATH

Name: _____ **Date:** _____

Part I • Key Concepts (Notes)

1. Exponents & Order of Operations

An **exponent** tells you how many times to multiply the base by itself.

$$3^4 = 3 \cdot 3 \cdot 3 \cdot 3 = 81, \qquad \left(\tfrac{2}{3}\right)^3 = \tfrac{2}{3} \cdot \tfrac{2}{3} \cdot \tfrac{2}{3} = \tfrac{8}{27}.$$

Order of Operations (PEMDAS). Work in this order: **P**arentheses → **E**xponents → **M**ultiply/**D**ivide (left to right) → **A**dd/**S**ubtract (left to right).

Example. $\dfrac{4+3^4}{5} = \dfrac{4+81}{5} = \dfrac{85}{5} = \boxed{17}$

Notice that grouping symbols come first: a fraction bar groups the whole numerator, so we evaluate $4+81$ *before* dividing by 5.

2. Square Roots & Cube Roots

Taking a root "undoes" a power. To solve an equation with an unknown power, ask: *what number, raised to that power, gives this result?*

$$\square^3 = 27 \implies \square = \sqrt[3]{27} = 3, \qquad \square^2 = 16 \implies \square = \pm 4.$$

Watch the sign. Every positive number has *two* square roots. $\sqrt{16} = 4$ means the **principal (positive)** root, while $-\sqrt{16} = -4$. So the full solution of $x^2 = 16$ is $x = \pm 4$. Cube roots keep the sign: $\sqrt[3]{27} = 3$ and $\sqrt[3]{-27} = -3$.

Check with squaring. $\sqrt{16} = 4$ because $4 \cdot 4 = 16$, i.e. $\sqrt{16} \cdot \sqrt{16} = 16$. Likewise $\left(\sqrt{17}\right)^2 = 17$ and $\left(\tfrac{13}{3}\right)^2 = \dfrac{169}{9}$, so $4\tfrac{1}{3} = \tfrac{13}{3}$ is the square root of $\tfrac{169}{9}$.

Estimating. $\sqrt{17}$ is between 4 and 5 because $4^2 = 16 < 17 < 25 = 5^2$; since 17 is close to 16, $\sqrt{17} \approx 4.1$.

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3. Converting Repeating Decimals to Fractions

Every decimal is one of three types.

(1) Terminating (e.g. 0.25) $\to$ rational

(2) Repeating (e.g. $0.\overline{2} = 0.222\ldots$) $\to$ rational

(3) Non-repeating, non-terminating (e.g. π) $\to$ **irrational**

Algebraic method (the reliable way). Multiply by a power of 10 so the repeating tails line up, then subtract.

Single repeating digit: convert $0.\overline{2}$.

$$\begin{aligned} x &= 0.222\ldots \\ 10x &= 2.222\ldots \\ 10x - x = 2 \implies 9x = 2 &\implies \boxed{x = \tfrac{2}{9}} \end{aligned}$$

Two repeating digits: convert $0.\overline{23}$. Multiply by 100.

$$\begin{aligned} x &= 0.2323\ldots \\ 100x &= 23.2323\ldots = 23 + x \\ 100x - x = 23 \implies 99x = 23 &\implies \boxed{x = \tfrac{23}{99}} \end{aligned}$$

Rule of thumb: for a purely repeating block of n digits, the denominator is n nines. $0.\overline{d} = \frac{d}{9}$, $0.\overline{ab} = \frac{ab}{99}$, $0.\overline{abc} = \frac{abc}{999}$. Always reduce to lowest terms.

4. Classifying Real Numbers

Think of the number sets as nested "bags": each set sits inside the larger one.

$$\underbrace{\mathbb{N}}_{\text{Natural}} \subset \underbrace{\mathbb{W}}_{\text{Whole}} \subset \underbrace{\mathbb{Z}}_{\text{Integer}} \subset \underbrace{\mathbb{Q}}_{\text{Rational}}, \qquad \text{Irrational} = \text{Real} \setminus \mathbb{Q}.$$

Quick definitions. **Natural** $= \{1, 2, 3, \ldots\}$ • **Whole** $= \{0, 1, 2, \ldots\}$ • **Integer** $= \{\ldots, -2, -1, 0, 1, 2, \ldots\}$ • **Rational** = any $\frac{a}{b}$ (terminating or repeating decimal) • **Irrational** = non-terminating, non-repeating (e.g. $\sqrt{12}$, π).

Number	Natural	Whole	Integer	Rational	Irrational
$\sqrt{9} = 3$	✓	✓	✓	✓	
$\frac{3}{10}$				✓	
-0.45				✓	
$\sqrt{12}$					✓

Tip: simplify first! $\sqrt{9} = 3$ is a natural number, but $\sqrt{12} = 2\sqrt{3} \approx 3.464\ldots$ never terminates or repeats, so it is irrational.

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Part II • Practice Problems

Show all work in the space provided.

Topic 1 • Order of Operations & Exponents

Evaluate each expression. Show every step.

1. EASY Evaluate 2^4.

2. EASY Evaluate $\left(\frac{3}{4}\right)^2$.

3. EASY Evaluate 5^3.

4. MEDIUM Evaluate $3^2 + 4^2$.

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5. MEDIUM Evaluate $(12-5)\cdot 2^3$.

6. MEDIUM Evaluate $\dfrac{4+81}{5}$.

7. MEDIUM Evaluate $0.5\times 0.5\times 0.5$.

8. HARD Evaluate $2\cdot 3^3-(7-3)^2$.

9. HARD Evaluate $\left(\dfrac{2}{3}\right)^3+\left(\dfrac{1}{2}\right)^2$.

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10. HARD Evaluate $\left[(5^2-1)\div 4\right]^2$.

Topic 2 • Square Roots & Cube Roots

Find each root or solution. Give all valid answers.

1. EASY Evaluate $\sqrt{49}$.

2. EASY Evaluate $\sqrt[3]{8}$.

3. EASY Evaluate $-\sqrt{25}$.

4. MEDIUM Solve $x^2 = 36$ (find both solutions).

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5. MEDIUM Solve $x^3 = 64$.

6. MEDIUM Evaluate $\left(\sqrt{13}\right)^2$.

7. MEDIUM Evaluate $\sqrt{\frac{9}{16}}$.

8. HARD Between which two consecutive integers does $\sqrt{17}$ lie? Explain how you know.

9. HARD Evaluate $\sqrt[3]{-27}$.

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10. HARD Solve $x^2 = 20$ and write the answer in simplest radical form.

Topic 3 • Repeating Decimals to Fractions

Convert each repeating decimal to a fraction in lowest terms. Use the algebraic method.

1. EASY Convert $0.\overline{5}$.

2. EASY Convert $0.\overline{7}$.

3. EASY Convert $0.\overline{1}$.

4. MEDIUM Convert $0.\overline{45}$.

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5. MEDIUM Convert $0.\overline{27}$.

6. MEDIUM Convert $0.\overline{8}$.

7. MEDIUM Convert $0.\overline{12}$.

8. HARD Convert $0.1\overline{6}$ (only the 6 repeats).

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9. HARD Convert $0.\overline{456}$.

10. HARD Convert $0.58\overline{3}$ (only the 3 repeats).

Topic 4 • Classifying Real Numbers

List every *set each number belongs to: Natural, Whole, Integer, Rational, Irrational. Simplify first.*

1. EASY Classify $\sqrt{9}$.

2. EASY Classify -7.

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3. EASY Classify $\frac{3}{10}$.

4. MEDIUM Classify $\sqrt{12}$.

5. MEDIUM Classify -0.45.

6. MEDIUM Classify 0.

7. MEDIUM Classify π.

8. HARD Classify $\sqrt{\frac{16}{25}}$.

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9. HARD Classify $0.1011011101111\ldots$ (pattern continues, never repeats a fixed block).

10. HARD Classify $\sqrt[3]{27}$.

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Part III • Detailed Answer Key

For teacher / self-check use. Each answer shows the reasoning, not just the result.

Topic 1 • Order of Operations & Exponents

1. $2^4 = 2 \cdot 2 \cdot 2 \cdot 2 = \mathbf{16}$.
2. $\left(\frac{3}{4}\right)^2 = \frac{3^2}{4^2} = \mathbf{\frac{9}{16}}$.
3. $5^3 = 5 \cdot 5 \cdot 5 = \mathbf{125}$.
4. Exponents first: $3^2 + 4^2 = 9 + 16 = \mathbf{25}$.
5. Parentheses, then exponent, then multiply: $(12 - 5) \cdot 2^3 = 7 \cdot 8 = \mathbf{56}$.
6. The fraction bar groups the numerator: $\frac{4+81}{5} = \frac{85}{5} = \mathbf{17}$.
7. $0.5 \times 0.5 = 0.25$, then $0.25 \times 0.5 = \mathbf{0.125}$ $\left(= \frac{1}{8}\right)$.
8. Exponents first: $2 \cdot 3^3 - (7 - 3)^2 = 2 \cdot 27 - 4^2 = 54 - 16 = \mathbf{38}$.
9. $\left(\frac{2}{3}\right)^3 + \left(\frac{1}{2}\right)^2 = \frac{8}{27} + \frac{1}{4} = \frac{32}{108} + \frac{27}{108} = \mathbf{\frac{59}{108}}$.
10. Inside brackets first: $5^2 - 1 = 24$, then $24 \div 4 = 6$, then $6^2 = \mathbf{36}$.

Topic 2 • Square Roots & Cube Roots

1. $\sqrt{49} = \mathbf{7}$ (since $7^2 = 49$).
2. $\sqrt[3]{8} = \mathbf{2}$ (since $2^3 = 8$).
3. $-\sqrt{25} = \mathbf{-5}$ (the negative of the principal root 5).
4. $x^2 = 36 \Rightarrow x = \pm\sqrt{36} = \pm\mathbf{6}$. Both 6 and -6 square to 36.
5. $x^3 = 64 \Rightarrow x = \sqrt[3]{64} = \mathbf{4}$ (cube roots give one real answer).
6. $\left(\sqrt{13}\right)^2 = \mathbf{13}$ (squaring undoes the square root).
7. $\sqrt{\frac{9}{16}} = \frac{\sqrt{9}}{\sqrt{16}} = \mathbf{\frac{3}{4}}$.
8. $\sqrt{17}$ lies between **4** and **5**, because $4^2 = 16 < 17 < 25 = 5^2$. Since 17 is nearer 16, $\sqrt{17} \approx 4.1$.
9. $\sqrt[3]{-27} = \mathbf{-3}$ (since $(-3)^3 = -27$; cube roots keep the sign).
10. $x^2 = 20 \Rightarrow x = \pm\sqrt{20} = \pm\sqrt{4 \cdot 5} = \pm\mathbf{2}\sqrt{\mathbf{5}}$.

Topic 3 • Repeating Decimals to Fractions

Same four steps every time: (1) let x = the decimal, (2) multiply by a power of 10 to shift, (3) subtract to erase the repeating tail, (4) solve and reduce.

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1. $x = 0.\overline{5}$; $10x = 5.\overline{5}$; $10x - x = 5 \Rightarrow 9x = 5 \Rightarrow x = \mathbf{\frac{5}{9}}$.
2. $x = 0.\overline{7}$; $10x = 7.\overline{7}$; $10x - x = 7 \Rightarrow 9x = 7 \Rightarrow x = \mathbf{\frac{7}{9}}$.
3. $x = 0.\overline{1}$; $10x = 1.\overline{1}$; $10x - x = 1 \Rightarrow 9x = 1 \Rightarrow x = \mathbf{\frac{1}{9}}$.
4. $x = 0.\overline{45}$; $100x = 45.\overline{45}$; $100x - x = 45 \Rightarrow 99x = 45 \Rightarrow x = \frac{45}{99} = \mathbf{\frac{5}{11}}$.
5. $x = 0.\overline{27}$; $100x = 27.\overline{27}$; $100x - x = 27 \Rightarrow 99x = 27 \Rightarrow x = \frac{27}{99} = \mathbf{\frac{3}{11}}$.
6. $x = 0.\overline{8}$; $10x = 8.\overline{8}$; $10x - x = 8 \Rightarrow 9x = 8 \Rightarrow x = \mathbf{\frac{8}{9}}$.
7. $x = 0.\overline{12}$; $100x = 12.\overline{12}$; $100x - x = 12 \Rightarrow 99x = 12 \Rightarrow x = \frac{12}{99} = \mathbf{\frac{4}{33}}$.
8. $x = 0.1\overline{6}$; $10x = 1.\overline{6}$; $100x = 16.\overline{6}$; $100x - 10x = 15 \Rightarrow 90x = 15 \Rightarrow x = \frac{15}{90} = \mathbf{\frac{1}{6}}$. *(Two shifts are needed because of the non-repeating "1" after the point.)*
9. $x = 0.\overline{456}$; $1000x = 456.\overline{456}$; $1000x - x = 456 \Rightarrow 999x = 456 \Rightarrow x = \frac{456}{999} = \mathbf{\frac{152}{333}}$.
10. $x = 0.58\overline{3}$; $100x = 58.\overline{3}$; $1000x = 583.\overline{3}$; $1000x - 100x = 525 \Rightarrow 900x = 525 \Rightarrow x = \frac{525}{900} = \mathbf{\frac{7}{12}}$. *(Two shifts because of the non-repeating "58".)*

Topic 4 • Classifying Real Numbers

1. $\sqrt{9} = 3$: **Natural, Whole, Integer, Rational**.
2. -7: **Integer, Rational** (negative, so not natural or whole).
3. $\frac{3}{10} = 0.3$: **Rational** only (a terminating decimal, not an integer).
4. $\sqrt{12} = 2\sqrt{3} \approx 3.464\ldots$: **Irrational** (not a perfect square).
5. -0.45: **Rational** only (terminating decimal $= -\frac{9}{20}$).
6. 0: **Whole, Integer, Rational** (not natural, since naturals start at 1).
7. $\pi \approx 3.14159\ldots$: **Irrational** (non-terminating, non-repeating).
8. $\sqrt{\frac{16}{25}} = \frac{4}{5}$: **Rational** only.
9. $0.1011011101111\ldots$: **Irrational** (never repeats a fixed block).
10. $\sqrt[3]{27} = 3$: **Natural, Whole, Integer, Rational**.

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